Q.25 Draw and explain the waveforms of a single-phase fully controlled bridge rectifier.

Q.26 Explain the working of a series inverter with circuit diagram and waveforms.

Q.27 Describe the types of chopper with their applications.

Q.28 Explain the working principle of cycloconverter and what are its applications?

Q.29 Explain the working of single phase full wave converter drive.

Q.30 Draw and explain the circuit diagram of step up chopper.

Q.31 Write short note on "Dual converter".

Q.32 Define Electric drives. Write any two disadvantages of AC drive over DC drive.

Q.33 What are the different types of batteries used in UPS system? Discuss their classification.

Q.34 Explain the construction and working principle of TRIAC with their V-I characteristics.

Q.35 How a GTO is different from an SCR? Explain the comparison of SCR and GTO.

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x10=20)

Q.36 Explain different methods of SCR triggering.

Q.37 Explain the different types of UPS systems. Discuss the working principle of each type and compare their advantages and disadvantages.

Q.38 Explain with the help of suitable diagram for automatically control the speed of a DC shunt motor.

Roll No. ....

4th Sem / Eltx

Subject:- Power Electronics

Time : 3Hrs. M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

Q.1 A DIAC is used to:

a) Commutate a TRIAC
b) Trigger a TRIAC
c) To protect a TRIAC
d) All of the above

Q.2 The primary functions of an SCR in a power electronics circuit:

a) Voltage control b) Current control
c) Power amplification d) Switching device

Q.3 UJT is a :

a) Current controlled device
b) Voltage controlled device
c) A relaxation oscillator
d) All of the above

Q.4 A fully controlled rectifier circuit contains:

a) Only SCRs
b) Only diodes
c) Mixture of SCRs and diodes
d) None of the above

Q.5 The duty cycle of a chopper is:
a) Ton/Toff b) Toff/Ton
c) Ton / (Ton + Toff) d) Toff/ (Ton + Toff)

Q.6 Cycloconveter drives are generally employed in;
a) Traction
b) Millling
c) Generating low frequencies
d) Generating pulses

Q.7 The TRIAC is equivalent to:
a) Two SCRs connected in parallel
b) Two SCRs connected in anti parallel
c) One SCR, one diode connected in parallel
d) One diode, one SCR connected in anti parallel

Q.8 In AC drive control, the purpose of constant V/F operation is:
a) To maintain constant voltage
b) To maintain a constant frequency
c) To provide smooth control of motor speed
d) To achieve high efficiency

Q.9 A UPS consists of:
a) Battery charger b) Inverter
c) Battery d) All of the above

Q.10 High-voltage DC transmission is used because:
a) It is costlier
b) It reduces line losses
c) It requires more transmission towers
d) It reduces power efficiency

SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

Q.11 A UJT has_______ PN junction.

Q.12 The turn on time of an SCR is about 2 to 5 μs (True/False)

Q.13 To turn off an SCR, it is necessary to reduce its current to less than trigger current. (True/False)

Q.14 The output voltage of a controlled rectifier is controlled by controlling ______ of thyristor.

Q.15 Lead acid batteries are costlier than SMF batteries. (True/False)

Q.16 Class-D chopper operates in_____ quadrants.

Q.17 A cycloconverter consist of______ and____ group of thyristors.

Q.18 In a dual converter one converter operates as ___________ and other______.

Q.19 In ON-line UPS, the inverter is_____ all the time.

Q.20 UPS is never used in street lighting. (True/False)

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

Q.21 Explain the working of an of SCR with the help of two transistor analogy.

Q.22 Write any five V-I characteristics of UJT.

Q.23 Define and explain the construction and working principle of DIAC.

Q.24 Explain the operation of single phase half-wave controlled rectifier with waveforms.